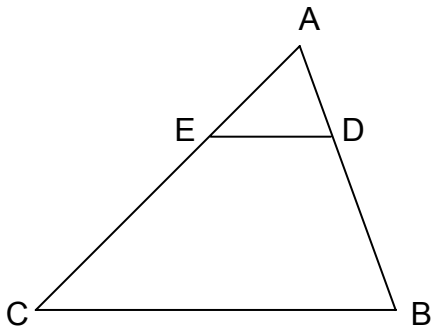


SAMPLE QUESTION PAPER

CLASS IX

MATHEMATICS

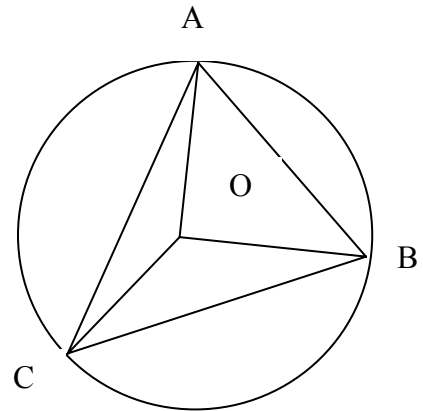
1. Find the cube root of 91125.
2. Evaluate: $\left(\frac{8}{27}\right)^{\frac{2}{3}} \times \left(\frac{27}{125}\right)^{\frac{1}{3}}$.
3. Using identity $x^2 - y^2 = (x + y)(x - y)$ find the product of 99×101 .
4. Expand $(2x + y + 3z)^2$
5. In figure, $DE \parallel BC$. If $AD = 3$ cm, $DB = 6$ cm and $EC = 8$ cm, find AE .



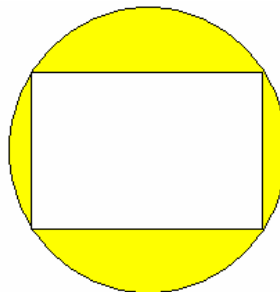
6. Factorise $x^2 - 5x - 6$.
7. Solve for x : $\frac{\frac{2}{3}x + 1}{x + \frac{1}{4}} = \frac{5}{3}$.
8. Find the circumference of a circle whose diameter is equal to 14 cm.
9. Find the mean value of the following data 45,34,36,42,40,41.
10. Find the area of an equilateral triangle of sides 20 cm each.
11. The two sides of a parallelogram are 20 cm and 25 cm. If the altitude corresponding to the sides of length 25 cm is 10 cm, find the altitude corresponding to the other pair of sides?

SAMPLE QUESTION PAPER

12. ΔABC is inscribed in a circle with centre O. Further $\angle AOB = 120^\circ$ and $\angle BOC = 150^\circ$ Find (i) $\angle BAC$ (ii) $\angle ACB$ (iii) $\angle ABC$.



13. The height and the slant height of a cone are 21 cm and 28 cm respectively. Find the volume of the cone?
14. The difference between two positive integers is 36. When one integer is divided by the other, the quotient is 4. Find the two integers.
15. Verify that $(2x + 3)$ is a factor of the polynomial $2x^2 + 5x + 3$.
16. A shirt was sold for Rs.442 after allowing a discount of 15% on the marked price. Find the marked price of the shirt.
17. Divide $3x^2 - 7x + 5$ by $(x - 2)$.
18. Factorize the following expression: $27x^3 - \frac{1}{216} - \frac{9}{2}x^2 + \frac{1}{4}x$.
19. A certain sum amounts to Rs.5832 in 2 years at 8% compound interest. Find the sum.
20. Find the area of the shaded region, in the figure given below, if the length of the rectangle is 4 cm and breadth is 3 cm.



ALL THE BEST